

Peptide QSAR Prediction Tool - Documentation

Purpose

This Windows-friendly local Streamlit application supports peptide-focused QSAR workflows: input validation, descriptor calculation, model training, prediction, interpretation, PCA, design suggestions, and export.

Developer

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Portfolio: <https://sites.google.com/view/ahmed-g-soliman/home>

Developer profile summary

Ahmed G. Soliman is presented on his portfolio as a MEXT master's student at Kyutech, School of Life Science and Engineering, with a biotechnology background from Ain Shams University, and CEO of iMole Bioinformatics Lab.

Workflow

1. Upload Data

Manual sequences, FASTA, CSV, Excel, TXT, and SDF files can be loaded. Sequence validation checks one-letter amino acid codes and reports invalid entries and duplicates.

2. Calculate Descriptors

The application computes peptide-specific descriptors and preserves uploaded activity/property columns.

3. Train Models

Regression, binary classification, and multiclass classification are supported. The user chooses target columns such as Activity, IC50, MIC, BindingScore, DockingScore, Vina, Affinity, Toxicity, or Solubility.

4. Evaluate, Predict, and Export

The app provides metrics, plots, prediction ranking, structure-quality criteria, model saving/loading, Excel/CSV/HTML exports, and this PDF documentation.

Peptide Descriptor Basis

Biopython ProteinAnalysis / ProtParam-style descriptors

Molecular weight, aromaticity, GRAVY, charge at pH 7, isoelectric point, instability index, secondary-structure fractions, and extinction coefficients.

Custom descriptors

Amino acid composition, dipeptide composition, residue class fractions, motif fingerprints, elemental composition, Shannon entropy, aliphatic index, Boman-style index, charge density, hydrophobic moment, and Chou-Fasman helix/sheet propensity averages.

Design criteria

The app adds heuristic peptide quality criteria for early screening. These are guidance values, not experimental proof.

Modeling and Evaluation

Models

Linear Regression, Ridge, Lasso, Elastic Net, Logistic Regression, SVR/SVM, kNN, Random Forest, Extra Trees, Gradient Boosting, Naive Bayes, and MLP. Optional packages can add XGBoost, LightGBM, SHAP, and UMAP.

Regression metrics

R², RMSE, MAE, cross-validation summary, actual-vs-predicted, and residual plots. Negative R² means the model performs worse than predicting the test-set mean.

Classification metrics

Accuracy, precision, recall, F1, ROC-AUC for binary tasks, classification report, and confusion matrix.

Position-Level Design Suggestions

The software can analyze residue-position effects against a numeric target. For each peptide position, it estimates which residues are associated with higher or lower target values.

For targets where higher is better, such as Activity or BindingScore, the design score favors larger target values. For targets where lower is usually better, such as IC50, MIC, DockingScore, Vina, or Affinity, the design score reverses direction.

Outputs include position-residue heatmaps, residue recommendations by position, residue-class suggestions, and mutation ideas for new peptides.

SDF Input Notes

SDF support is intended for docking or cheminformatics exports that include peptide sequence as a property. Supported property names include Sequence, Peptide Sequence, Peptide_Sequence, AA Sequence, amino acid sequence, and similar variants.

The app also attempts value-based detection when a property contains valid one-letter amino acid sequences. If an SDF file only contains coordinates and no peptide sequence property, peptide descriptors cannot be calculated from it.

Documentation Sources Used

Streamlit documentation: <https://docs.streamlit.io/>

Biopython ProtParam documentation: <https://biopython.org/wiki/ProtParam>

scikit-learn user guide: https://scikit-learn.org/stable/user_guide.html

pandas documentation: <https://pandas.pydata.org/docs/>

Plotly Python documentation: <https://plotly.com/python/>

SHAP documentation: <https://shap.readthedocs.io/>

PyInstaller documentation: <https://pyinstaller.org/en/stable/>

Scientific formulas and scales implemented manually include Kyte-Doolittle hydrophobicity, Chou-Fasman propensities, aliphatic index, Boman-style residue scale, and hydrophobic moment approximation.

Packaging

Normal Windows launch
Double-click run_app.bat.

Single Python file

Use peptide_qsar_single_file.py for a self-contained source launcher that writes the embedded app files to a runtime folder and starts Streamlit.

PyInstaller example

```
pyinstaller --noconfirm --onefile --name PeptideQSAR peptide_qsar_single_file.py --collect-all streamlit --collect-all pandas --collect-all numpy --collect-all sklearn --collect-all Bio --collect-all plotly --collect-all scipy --collect-all openpyxl
```